TEAS 7 Science Review Sheets

Review of the four TEAS 7 Science sub-areas: anatomy & physiology, biology, chemistry, scientific reasoning

MindShelf

The Science Section at a Glance

Sub-area (ATI TEAS 7 content outline)	Scored items
Human anatomy & physiology: Body orientation; respiratory, cardiovascular, digestive, nervous, muscular, skeletal, reproductive, integumentary, endocrine, urinary and immune systems	18
Biology: Cells; DNA, RNA and proteins; Mendelian inheritance; macromolecules; microorganisms and disease	9
Chemistry: Atomic structure; properties and changes of matter; reactions and reaction conditions; solutions; acids and bases	8
Scientific reasoning: Measurements and tools; logic and evidence; relationships between variables; the scientific method	9

44 scored items plus 6 unscored pretest items (50 in all) in 60 min. Pretest items are mixed in with the scored ones and do not count, so give every question full effort. These sheets review core facts for each sub-area; they are not a complete course and contain no actual test questions.

Human Anatomy & Physiology

Largest sub-area: 18 scored items.

Orientation

Anatomical position: standing upright, facing forward, arms at the sides, palms forward, feet parallel. **Supine** = lying face up; **prone** = face down. Left and right always mean the *patient's* left and right.

Term	Opposite	Meaning
Superior (cranial)	Inferior (caudal)	Toward the head / toward the feet
Anterior (ventral)	Posterior (dorsal)	Front / back of the body
Medial	Lateral	Toward / away from the midline
Proximal	Distal	Closer to / farther from the trunk or a limb's attachment
Superficial	Deep	Toward / away from the body surface

Plane	Divides the body into
Sagittal (midsagittal = equal halves)	Right and left parts
Frontal (coronal)	Anterior and posterior parts
Transverse (horizontal)	Superior and inferior parts

Cavities: dorsal = cranial (brain) + vertebral (spinal cord); ventral = thoracic (pleural cavities, pericardial cavity, mediastinum) + abdominopelvic, separated by the diaphragm.

Levels of organization: chemical → cell → tissue → organ → organ system → organism. Four tissue types: epithelial, connective, muscle, nervous.

Homeostasis: negative feedback reverses a change (body temperature near 37 °C / 98.6 °F, blood glucose, blood pH 7.35–7.45); positive feedback amplifies it until an end point (oxytocin in labor, blood clotting).

Cardiovascular System

Right atrium	Receives deoxygenated blood from the superior and inferior venae cavae
Right ventricle	Pumps blood through the pulmonary valve into the pulmonary trunk (pulmonary circuit)
Left atrium	Receives oxygenated blood from the pulmonary veins
Left ventricle	Thickest wall; pumps blood through the aortic valve into the aorta (systemic circuit)
Valves	Tricuspid (right) and mitral or bicuspid (left); semilunar valves (pulmonary and aortic) guard the outflow; all prevent backflow
Conduction	SA node (pacemaker) → AV node → AV bundle (bundle of His) → bundle branches → Purkinje fibers
Vessels	Arteries carry blood away from the heart; capillaries (one cell thick) exchange gases and nutrients; veins return blood, with valves in the limbs
Blood	Erythrocytes carry O_2 on hemoglobin; leukocytes defend; platelets start clotting; plasma carries solutes

4 chambers. Blood path: venae cavae → right atrium → tricuspid → right ventricle → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral → left ventricle → aorta → body.

Cardiac output

$$CO = HR \times SV$$

E.g. HR 70/min, SV 70 mL: CO = 70 × 70 mL/min = 4.9 L/min.

Type	RBC antigens	Plasma antibodies	Can receive red cells from
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A and B	none	A, B, AB, O (universal recipient)
O	none	anti-A and anti-B	O only (universal donor of red cells)

Rh: Rh-positive cells carry the D antigen. An Rh-negative person makes anti-D only after exposure (transfusion, pregnancy with an Rh-positive fetus), which can cause hemolytic disease of the newborn in a later pregnancy.

Respiratory System

Nasal cavity	Warms, humidifies and filters inhaled air
Pharynx and larynx	Shared airway and food passage; larynx holds the vocal cords; the epiglottis covers it during swallowing
Trachea and bronchi	Cartilage rings keep the trachea open; bronchi branch into bronchioles (conducting zone)
Alveoli	Gas exchange by diffusion across a thin respiratory membrane; surfactant lowers surface tension
Diaphragm	Contracts and flattens on inspiration: chest volume up, pressure down, air flows in
Brainstem centers	Medulla and pons set the breathing rhythm; rising CO_2 (falling pH) is the main drive to breathe

Right lung 3 lobes, left lung 2 lobes (room for the heart). O_2 and CO_2 move by diffusion, from higher to lower partial pressure.

Digestive System

Mouth	Chewing (mechanical digestion); salivary amylase starts starch digestion
Esophagus	Moves the bolus to the stomach by peristalsis
Stomach	HCl and pepsin begin protein digestion; churning makes chyme
Small intestine	Duodenum, jejunum, ileum: most chemical digestion and absorption; villi and microvilli enlarge the surface
Liver and gallbladder	Liver makes bile (emulsifies fats); gallbladder stores and concentrates it
Pancreas (exocrine)	Releases amylase, lipase, proteases and bicarbonate into the duodenum
Large intestine	Absorbs water and electrolytes; gut bacteria make vitamin K; forms feces

Order: mouth → pharynx → esophagus → stomach → small intestine → large intestine → rectum → anus. Liver, gallbladder and pancreas are accessory organs: food never passes through them.

Nervous System

CNS	Brain and spinal cord: integrate information
Neuron	Dendrites receive, the cell body integrates, the axon conducts; myelin (Schwann cells in the PNS, oligodendrocytes in the CNS) speeds conduction
Cerebrum	Lobes: frontal (motor, planning, personality), parietal (touch, body sense), temporal (hearing, memory), occipital (vision)
Cerebellum	Coordination, balance, posture
Brainstem	Midbrain, pons, medulla oblongata: breathing, heart rate, blood pressure
Autonomic NS	Sympathetic: fight or flight (heart rate up, pupils dilate, airways dilate, digestion slows); parasympathetic: rest and digest
Action potential	Na^+ rushes in (depolarization), K^+ flows out (repolarization); neurotransmitters such as acetylcholine cross the synapse

PNS: 12 pairs of cranial nerves, 31 pairs of spinal nerves; afferent (sensory) toward the CNS, efferent (motor) away from it.

Muscular & Skeletal Systems

Skeletal muscle	Voluntary, striated, multinucleated fibers
Cardiac muscle	Involuntary, striated, branched cells joined by intercalated discs
Smooth muscle	Involuntary, not striated; walls of hollow organs and blood vessels
Sarcomere	Contraction unit: actin (thin) and myosin (thick) filaments slide past each other; needs Ca^{2+} and ATP
Bone functions	Support, protection, movement, mineral (Ca^{2+}) storage, blood cell formation in red marrow
Bone cells	Osteoblasts build bone, osteoclasts break it down, osteocytes maintain it
Connections	Tendons join muscle to bone; ligaments join bone to bone; synovial joints move freely

Adult skeleton: 206 bones. Axial = skull, vertebral column, rib cage; appendicular = limbs and their girdles. Vertebrae: 7 cervical, 12 thoracic, 5 lumbar, then sacrum and coccyx.

Endocrine System

Hypothalamus	Releasing and inhibiting hormones control the anterior pituitary; makes ADH and oxytocin
Anterior pituitary	GH, TSH, ACTH, FSH, LH, prolactin
Posterior pituitary	Releases ADH (water reabsorption) and oxytocin (uterine contraction, milk ejection)
Thyroid	T_4 and T_3 raise metabolic rate; calcitonin lowers blood Ca^{2+}
Parathyroids	PTH raises blood Ca^{2+}
Adrenal cortex	Cortisol (stress, raises glucose); aldosterone (Na^+ and water retention)
Adrenal medulla	Epinephrine and norepinephrine (fight or flight)

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Pancreatic islets	Insulin from beta cells lowers blood glucose; glucagon from alpha cells raises it
Pineal gland	Melatonin (sleep–wake rhythm)

Most hormone levels are held steady by negative feedback (e.g., high T_4 suppresses TSH). **Endocrine** glands are ductless and release hormones into the blood; **exocrine** glands secrete through ducts (sweat, salivary, the enzyme-secreting part of the pancreas).

Urinary System

Kidneys	Filter blood; regulate water, electrolytes, pH and blood pressure; release renin and erythropoietin (EPO)
Nephron	Glomerulus and glomerular (Bowman's) capsule filter → proximal tubule (most reabsorption) → nephron loop (loop of Henle) → distal tubule → collecting duct
Hormonal control	ADH increases water reabsorption in collecting ducts; aldosterone increases Na^+ reabsorption
Urine path	Ureters (peristalsis) → urinary bladder (storage) → urethra

Reproductive System

Testes	Sperm production in seminiferous tubules; testosterone from interstitial cells
Male ducts and glands	Epididymis (maturation, storage) → ductus (vas) deferens → urethra; seminal vesicles, prostate and bulbourethral glands add semen fluid
Ovaries	Oocytes; estrogen and progesterone
Uterine tubes and uterus	Fertilization usually occurs in the uterine (fallopian) tube; implantation in the endometrium, which sheds at menstruation
Hormones	FSH and LH from the anterior pituitary drive gamete production; the LH surge triggers ovulation

Integumentary System

Epidermis	Keratinized stratified squamous epithelium, no blood vessels; melanocytes make melanin (UV protection)
Dermis	Connective tissue with blood vessels, nerves, hair follicles, sweat and sebaceous glands
Hypodermis	Subcutaneous fat: insulation, energy storage, anchoring
Functions	Barrier, thermoregulation (sweat, blood flow), sensation, vitamin D synthesis

Immune & Lymphatic Systems

Innate defenses	Nonspecific: skin and mucous membranes, phagocytes (neutrophils, macrophages), NK cells, inflammation, fever
B cells	Humoral immunity: become plasma cells that secrete antibodies; mature in red bone marrow
T cells	Cell-mediated immunity: helper T cells coordinate, cytotoxic T cells kill infected cells; mature in the thymus
Lymphatic organs	Lymph nodes filter lymph; spleen filters blood and removes old red cells; lymphatic vessels return fluid to the blood
Immunity types	Active (infection or vaccine, makes memory cells) vs passive (antibodies received, e.g., across the placenta; short-lived)

Biology

9 scored items.

Cells & Organelles

Structure	Function
Nucleus	Holds the DNA; the nucleolus assembles ribosome subunits
Ribosome	Builds proteins (free in cytosol or bound to rough ER)
Rough ER	Ribosome-studded; makes proteins for membranes and export
Smooth ER	Makes lipids, detoxifies, stores Ca^{2+}
Golgi apparatus	Modifies, sorts and packages proteins into vesicles
Mitochondrion	Aerobic cellular respiration → ATP; has its own DNA
Lysosome	Digestive enzymes break down waste and worn-out parts (animal cells)
Cell membrane	Phospholipid bilayer with proteins (fluid mosaic), selectively permeable
Cytoskeleton	Microtubules and microfilaments: shape, transport, movement
Centrioles	Organize the spindle in animal cells
Plant cells only	Cell wall (cellulose), chloroplasts (photosynthesis), large central vacuole

Prokaryotes (bacteria, archaea): no nucleus, no membrane-bound organelles; DNA in a nucleoid; ribosomes present. **Eukaryotes** (animals, plants, fungi, protists): nucleus and organelles.

Membrane transport. Passive, no ATP, down the gradient: diffusion, facilitated diffusion, osmosis (water moves toward the higher solute concentration). Active, uses ATP, against the gradient: e.g., the Na^+/K^+ pump, endocytosis, exocytosis.

Cell placed in	Result
Hypertonic solution	Water leaves; the cell shrinks (crenates)
Hypotonic solution	Water enters; the cell swells, may burst (lyse)
Isotonic solution	No net water movement

Cellular respiration and photosynthesis

Aerobic respiration: $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + ATP$. Glycolysis (cytoplasm, no O_2 needed) → citric acid (Krebs) cycle (mitochondrial matrix) → electron transport chain (inner mitochondrial membrane; O_2 is the final electron acceptor; makes most of the ATP). Without O_2, cells rely on fermentation (lactic acid in human muscle).
Photosynthesis (chloroplasts): $6\ CO_2 + 6\ H_2O + light \rightarrow C_6H_{12}O_6 + 6\ O_2$. Its products are the reactants of respiration.

Cell Division

Cell cycle: interphase (G_1 growth, S DNA replication, G_2 preparation) → mitosis: prophase, metaphase (chromosomes line up), anaphase (sister chromatids separate), telophase (PMAT) → cytokinesis.

	Mitosis	Meiosis
Purpose	Growth, repair, asexual reproduction	Gametes (eggs and sperm)
Divisions	One	Two (meiosis I and II)
Daughter cells	Two, genetically identical	Four, genetically different
Chromosome number	Diploid (2n) kept	Haploid (n): halved
Variation	None (barring mutation)	Crossing over (prophase I) and independent assortment

Humans: 2n = 46 (23 pairs); a gamete has $n = 46/2 = 23$.

DNA, RNA & Protein Synthesis

	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Strands	Double helix	Usually single
Role	Stores genetic information	Carries it (mRNA), brings amino acids (tRNA), forms ribosomes (rRNA)

Base pairing: A–T and G–C in DNA; A–U in RNA. DNA replication is semiconservative (each new helix keeps one old strand).

Central dogma
DNA → **transcription** (nucleus, RNA polymerase) → mRNA → **translation** (ribosome, tRNA brings amino acids) → protein

A **codon** is 3 bases of mRNA; the tRNA **anticodon** pairs with it. $4^3 = 64$ codons code for 20 amino acids plus 3 stop codons (UAA, UAG, UGA): the code is redundant. AUG = start (methionine). *E.g.* template DNA TAC CGT → mRNA AUG GCA → Met–Ala.

Mutations: substitution (point; may change one amino acid, none, or create a stop codon); insertion or deletion (shifts the reading frame: frameshift).

Mendelian Genetics

Allele: version of a gene. **Genotype** (AA, Aa, aa) vs **phenotype** (observed trait). Homozygous = two identical alleles; heterozygous = two different. **Law of segregation:** allele pairs separate into gametes. **Independent assortment:** genes on different chromosomes sort independently.

EXAMPLE Punnett square, Aa × Aa

	A	a
A	AA	Aa
a	Aa	aa

Genotypes AA : Aa : aa = 1 : 2 : 1 phenotypes dominant : recessive = 3 : 1 P(aa) = 0.5 × 0.5 = 25 %.
- Test cross Aa × aa: P(aa) = 0.5 × 1 = 50 %.
- Dihybrid AaBb × AaBb, out of 16: 9 : 3 : 3 : 1 (e.g., both dominant = 0.75 × 0.75 × 16 = 9).

Non-Mendelian: incomplete dominance (blend: red × white → pink); codominance (both expressed: blood type AB); multiple alleles (ABO); X-linked recessive traits (e.g., hemophilia) appear more often in males.

Macromolecules & Enzymes

Class	Building block	Examples / role
Carbohydrates	Monosaccharides (e.g. glucose)	Glucose, sucrose, starch, glycogen, cellulose. Quick energy, storage, plant structure
Lipids	Glycerol + fatty acids (not true polymers)	Triglycerides, phospholipids, steroids (cholesterol). Long-term energy, membranes, hormones, insulation
Proteins	Amino acids joined by peptide bonds	Enzymes, hemoglobin, collagen, antibodies. Catalysis, structure, transport, defense; shape sets function
Nucleic acids	Nucleotides (sugar + phosphate + base)	DNA, RNA (ATP is a nucleotide). Store and express genetic information

Dehydration synthesis joins monomers (releases water); hydrolysis splits them (uses water). **Enzymes** are mostly proteins: they lower activation

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energy, fit specific substrates at the active site and are not used up; heat and extreme pH denature them.

Microorganisms & Disease

Group	Structure	Examples / treatment
Bacteria	Prokaryotic cells; peptidoglycan wall; cocci, bacilli, spirilla	Strep throat, tuberculosis; treated with antibiotics
Viruses	Not cells: DNA or RNA in a protein capsid; replicate only inside host cells	Influenza, HIV; antibiotics do not work; antivirals, vaccines
Fungi	Eukaryotic; chitin cell wall; yeasts and molds	Candidiasis, athlete's foot; antifungals
Protozoa	Single-celled eukaryotes	Malaria (Plasmodium), giardiasis

Gram stain: Gram-positive = thick peptidoglycan wall, stains purple; Gram-negative = thin wall plus outer membrane, stains pink.

Pathogens cause disease by invading tissue or releasing toxins; normal flora (e.g., gut bacteria) protect the host and help digestion. Hand hygiene breaks the chain of infection between reservoir and susceptible host.

Transmission	How	Example
Direct contact	Touching, kissing, sexual contact, contact with soil or vegetation	Skin infections
Droplet spread	Large respiratory droplets sprayed over a short distance (sneezing, coughing, talking)	Pertussis, influenza
Airborne	Small particles that stay suspended in air, carried by dust or air currents	Pulmonary tuberculosis
Vehicle-borne	Contaminated food, water, blood products or objects (fomites)	Hepatitis B via blood
Vector-borne	Carried by a living vector, usually an insect	Malaria (mosquito)

Chemistry

8 scored items.

Atomic Structure & Periodic Table

Particle	Charge	Location
Proton	positive	nucleus; their number is the atomic number Z and defines the element
Neutron	none	nucleus
Electron	negative	shells (energy levels) around the nucleus; valence electrons drive bonding

Mass number $A = Z + N$; **isotopes** = same Z, different N; **ion** = atom that gained (anion, −) or lost (cation, +) electrons. *E.g.* carbon-14: $N = 14 - 6 = 8$; chlorine-35: $N = 35 - 17 = 18$; Na^+: 11 protons − 10 electrons = charge +1.

Periodic table: rows (periods) = number of electron shells; columns (groups) = same valence electrons, similar properties. Metals on the left, nonmetals on the right, metalloids along the staircase; alkali metals (far-left column) are very reactive; noble gases (far-right column) are stable and unreactive. Electronegativity increases toward the upper right (fluorine highest); atomic radius increases toward the lower left.

Chemical Bonds

Bond	How	Example
Ionic	Electrons transferred (metal → nonmetal); oppositely charged ions attract	NaCl
Nonpolar covalent	Electrons shared equally	H_2, O_2
Polar covalent	Electrons shared unequally: partial charges (δ+ / δ−)	H_2O
Hydrogen bond	Weak attraction between H bonded to N/O/F and a nearby N/O/F	Between water molecules, DNA base pairs
Metallic	Cations in a sea of shared electrons	Cu, Fe

Properties & States of Matter

Solid: fixed shape and volume. **Liquid:** fixed volume, takes the container's shape. **Gas:** neither; compressible.

Phase change	From	To	Heat
Melting	solid	liquid	absorbed
Freezing	liquid	solid	released
Vaporization (evaporation / boiling)	liquid	gas	absorbed
Condensation	gas	liquid	released
Sublimation	solid	gas	absorbed
Deposition	gas	solid	released

Density = m/V (a physical property): 50 g occupying 20 mL = 2.5 g/mL; a less dense object floats on a denser liquid.

Physical change: same substance (melting, dissolving, cutting). **Chemical change:** new substance (signs: gas, color change, precipitate, heat or light). Water melts at 0 °C and boils at 100 °C (1 atm); ice floats because hydrogen bonding makes it less dense than liquid water; water has a high specific heat, cohesion and adhesion.

Gas laws (temperature in kelvins)

$$P_1V_1 = P_2V_2 \quad \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Boyle (constant T): pressure up, volume down; this is how breathing moves air. Charles (constant P): heat a gas, it expands. Gay-Lussac (constant V): heat a gas, its pressure rises. *E.g.* 6 L at 1 atm squeezed to 2 atm: $V_2 = 1 \times 6/2 = 3$ L.

Chemical Reactions

Type	Pattern	Example
Synthesis	A + B → AB	$2\,H_2 + O_2 \rightarrow 2\,H_2O$
Decomposition	AB → A + B	$2\,H_2O_2 \rightarrow 2\,H_2O + O_2$
Single replacement	A + BC → AC + B	$Zn + 2\,HCl \rightarrow ZnCl_2 + H_2$
Double replacement	AB + CD → AD + CB	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
Combustion	fuel + O_2 → CO_2 + H_2O	$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$
Neutralization	acid + base → salt + water	$HCl + NaOH \rightarrow NaCl + H_2O$

Balancing: conservation of mass
Change coefficients only, never subscripts. $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$: O left $2 \times 2 = 4$, O right $2 + 2 = 4$; H right $2 \times 2 = 4$, as in CH_4.

Faster reactions: higher temperature, higher concentration, more surface area, a catalyst (lowers activation energy, not consumed). **Exothermic** releases heat; **endothermic** absorbs it. **Le Chatelier:** a system at equilibrium shifts to offset a change in concentration, temperature or pressure.

Solutions

Solute dissolves in solvent (water: the "universal solvent"); like dissolves like (polar with polar). Unsaturated → saturated → supersaturated. Most solids dissolve better when hot; gases dissolve better when cold and under pressure.

Concentration

$$M = \frac{\text{mol solute}}{\text{L solution}} \quad C_1V_1 = C_2V_2 \quad \%\,(m/v) = \frac{\text{g solute}}{100\ \text{mL solution}}$$

- NaCl molar mass 22.99 + 35.45 = 58.44 g/mol; 14.61 g in 0.5 L: $M = 14.61/(58.44 \times 0.5) = 0.50$ M.
- 500 mL of 0.5 M from 2 M stock: $V_1 = 0.5 \times 500/2 = 125$ mL, then add solvent to 500 mL.
- 250 mL of 5 % (m/v) glucose holds $5 \times 250/100 = 12.5$ g.
- Moles $n = m/M$: $H_2O = 2 \times 1.008 + 15.999 \approx 18.0$ g/mol; 36.03 g of water = 2.0 mol.

Acids & Bases

Acid: donates H^+, pH < 7, sour, turns blue litmus red. **Base:** accepts H^+ (or releases OH^-), pH > 7, bitter, slippery, turns red litmus blue. Neutral = 7 (25 °C); scale 0–14.

$$pH = -\log[H^+] \quad pH + pOH = 14 \quad (25\ °C)$$

Each pH unit is a tenfold change: pH 3 has $10^{6-3} = 1{,}000$ times the $[H^+]$ of pH 6. At pH 3, pOH = 14 − 3 = 11.

Strong acids (HCl) ionize fully; weak acids (acetic, carbonic) only partly. **Buffers** resist pH change: the bicarbonate buffer keeps blood pH at 7.35–7.45.

Scientific Reasoning

9 scored items.

Scientific Method

Observation → question → **hypothesis** (testable, falsifiable prediction) → controlled experiment → data analysis → conclusion → repeat and communicate.

Term	Meaning
Independent variable	What the investigator changes (x-axis)
Dependent variable	What is measured in response (y-axis)
Controlled variables	Kept the same in every trial
Control group	Gets no treatment: the baseline for comparison
Theory	Broad, well-tested explanation
Law	Describes what happens (often an equation), not why

Accuracy = closeness to the true value; **precision** = closeness of repeated results to each other. More trials and larger samples make results more reliable. Correlation does not prove causation. An **observation** is what you sense or measure; an **inference** is a conclusion drawn from it. Inductive reasoning goes from specific cases to a general rule; deductive reasoning applies a general rule to a specific case.

Working with data

- Mean of trials 12, 15, 18: $(12 + 15 + 18)/3 = 15$.
- Slope: 4 cm on day 2, 10 cm on day 8: $(10 - 4)/(8 - 2) = 1$ cm/day.
- Percent error: measured 10.4, accepted 10: $(10.4 - 10)/10 \times 100 = 4$ %.

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Judging an investigation

- Only one independent variable should change; everything else stays controlled.
- Look for a control group, enough subjects and repeated trials.
- A conclusion must stay within the data: no claims beyond the conditions tested.
- A hypothesis is supported or not supported, never "proven".
- Check graphs for title, labeled axes, units and scale before reading a trend.

Graphs: line = change over time or a continuous variable; bar = compare categories; pie = parts of a whole; scatter = relationship between two variables (positive, negative or none). Interpolation reads between data points; extrapolation goes beyond them (less reliable).

Measurement & Units

Tools: balance (mass), graduated cylinder (volume: read the bottom of the meniscus at eye level), ruler or meter stick (length), thermometer (temperature), pipette (small volumes). **Mass** (kg, same everywhere) is not **weight** (a force that depends on gravity).

Prefix	Symbol	× base unit
kilo	k	1,000
(base unit)	—	—
deci	d	0.1
centi	c	0.01
milli	m	0.001
micro	µ	0.000001

1 in = 2.54 cm · 1 ft = 12 in · 1 lb ≈ 0.4536 kg · 1 kg ≈ 2.2 lb · 1 L = 1,000 mL · 1 mL = 1 cm^3

Conversions

- 2.5 L = 2,500 mL · 250 mg = 0.25 g · 0.5 mg = 500 µg · 0.75 km = 750 m
- 150 lb × 0.45359237 ≈ 68.0 kg
- 5 ft 6 in = (5 × 12 + 6) in × 2.54 ≈ 167.6 cm
- °C = (°F − 32)/1.8: 98.6 °F = 37 °C; °F = 1.8 × °C + 32: 40 °C = 104 °F
- K = °C + 273.15: 37 °C = 310.15 K

Moving to a smaller unit gives a bigger number (multiply); moving to a larger unit gives a smaller number (divide).

Common Traps

⚠ TRAP

- Pulmonary arteries carry deoxygenated blood and pulmonary veins oxygenated blood: arteries are defined by direction (away from the heart), not by O_2.
- Tendons join muscle to bone; ligaments join bone to bone.
- Insulin lowers blood glucose; glucagon raises it. Calcitonin lowers blood Ca^{2+}; PTH raises it.
- Mitosis gives two identical diploid cells; meiosis gives four different haploid cells.
- RNA has uracil, not thymine; codons are read on mRNA.
- Aa × Aa gives a 3 : 1 phenotype ratio but a 1 : 2 : 1 genotype ratio.
- Antibiotics act on bacteria, not on viruses.
- Lower pH = more acidic; one pH unit = tenfold change in $[H^+]$.
- Balance with coefficients, never by changing subscripts.
- The independent variable is changed; the dependent variable is measured.
- Precise is not the same as accurate.
- Supine is face up; prone is face down.
- ADH saves water; aldosterone saves sodium (water follows).
- Food passes through the pharynx but never through the liver or pancreas.
- Gas-law calculations need kelvins, not degrees Celsius.

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